

AMATH 482/582: HOME WORK 1

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ABSTRACT. This study shows the path of the submarine moving in the Puget Sound while the acoustic pressure data were collected for over 24 hours in half-hour increments. Results indicate how effective it is to use Fourier transform and 3D Gaussian filter on the noisy data since they help find an accurate and clear path. Figures show that the plot with noise data has outliers, but after filtering the data, those outliers are removed, presenting the submarine's direct and smooth path. This study can be used to create a sub-tracking aircraft in the future to detect the location of a moving submarine.

1. Introduction and Overview

A submarine with a new technology has been developed. The submarine emits unknown acoustic frequencies that have to be detected to know the path and location of the submarine. However, one issue is that the detection of the submarine using the collected acoustic pressure measurement data is not accurate enough since it contains noisy data. By using the Fourier transform, the dominant frequency would be identified and that represents the most significant data of this new complex data. Also the 3D Gaussian filter would remove the noisy data and provide the denoised ones which is valid in detecting the submarine. This report aims to detect and get a valid path of the submarine by using the Fourier Transform and the 3D Gaussian filter on the acoustic pressure data.

2. Theoretical Background

The Discrete Fourier Transform transforms the discrete time domain signal (original collected data) into its frequency domain representation.

Discrete Fourier Transform formula in three dimensions are given by:

$$(1) \quad \hat{f}(\kappa_x, \kappa_y, \kappa_z) = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} \sum_{z=0}^{N_z-1} f(x, y, z) e^{-i2\pi(\frac{\kappa_x x}{N_x} + \frac{\kappa_y y}{N_y} + \frac{\kappa_z z}{N_z})}$$

and the Inverse of Discrete Fourier Transform formula in three dimensions are given by:

$$(2) \quad f(x, y, z) = \frac{1}{N_x N_y N_z} \sum_{\kappa_x=0}^{N_x-1} \sum_{\kappa_y=0}^{N_y-1} \sum_{\kappa_z=0}^{N_z-1} \hat{f}(\kappa_x, \kappa_y, \kappa_z) e^{i2\pi(\frac{\kappa_x x}{N_x} + \frac{\kappa_y y}{N_y} + \frac{\kappa_z z}{N_z})}$$

Fourier shift formula in three dimensions is given by:

$$(3) \quad f(x - x_0, y - y_0, z - z_0) \longleftrightarrow \hat{f}(\kappa_x, \kappa_y, \kappa_z) e^{-i(\kappa_x x_0 + \kappa_y y_0 + \kappa_z z_0)}$$

- $f(x, y, z)$ is the noisy original acoustic pressure measurement.
- $\hat{f}(x, y, z)$ is the Fourier Transform of $f(x, y, z)$ in frequency domain.
- N_x, N_y, N_z is 64, as the given data is in $64 \times 64 \times 64$ by each time.
- $(\kappa_x, \kappa_y, \kappa_z)$ is the discrete frequency component of its frequency domain representation of the 3D acoustic pressure measurement that corresponds to (x, y, z) .

- (x_0, y_0, z_0) is the spatial shift in the (x, y, z) direction.

The Discrete Fourier Transformation (1) is necessary for this study because the collected complex acoustic pressure data contains noise in addition to the signal we need to analyze. The Fourier Transform (1) helps identify the dominant frequency among the unknown acoustic frequencies, which aids in detecting the frequency of the submarine's acoustic signals. The Fourier Shift (3) repositions the Fourier transformed data by centering the DC component in the middle of the spectrum. This is crucial for visualizing and interpreting the signal data patterns as it organizes the frequency data to make tracking the submarine easier.

In Task 1, averaging of the Fourier transform is used for determining the dominant frequency. Below is the mathematical reason of how and why averaging of the Fourier transform qualifies in determining the dominant frequency.

$$(4) \quad Y_t = X_t + \epsilon$$

$$(5) \quad \hat{Y}_t = \hat{X}_t + \hat{\epsilon}$$

$$(6) \quad \mathbb{E}[\hat{Y}_t] = \mathbb{E}[\hat{X}_t + \hat{\epsilon}]$$

$$(7) \quad \mathbb{E}[\hat{Y}_t] = \mathbb{E}[\hat{X}_t] + \mathbb{E}[\hat{\epsilon}]$$

$$(8) \quad \mathbb{E}[\hat{Y}_t] = \mathbb{E}[\hat{X}_t]$$

- Y_t is the original acoustic pressure measurement that has been collected.
- X_t is the clean (denoised) data that is wanted in here to get a clear path of the submarine.
- ϵ is the noise in the collected data.
- $\mathbb{E}$ is the notation that means for averaging the value inside [].

Using Fourier transform, (4) would become (5) while (6) and (7) shows the averaging of the Fourier transform by time. The averaging of the Fourier transform noise ($\mathbb{E}[\hat{\epsilon}]$) would become 0, leading to the result of (8). (8) tells that by Fourier transforming and averaging (4), the noised data would get removed and this tells us that averaging the clean/denoised data in Fourier space would be the same to averaging original data that is in Fourier space. By using $\mathbb{E}[\hat{X}_t]$, the index of the dominant frequency could be found. This is the representative signal that is used to reduce the complexity of the data by filtering out the noise and focusing on the significance of the data.

In Task 2, the 3D Gaussian filter is needed to remove the noisy data and provide a clear path of the submarine with time. The equation of the 3D Gaussian filter is given by:

$$(9) \quad G(\kappa_x, \kappa_y, \kappa_z) = e^{-\tau((\kappa_x - \kappa_x^*)^2 + (\kappa_y - \kappa_y^*)^2 + (\kappa_z - \kappa_z^*)^2)}$$

- $(\kappa_x^*, \kappa_y^*, \kappa_z^*)$ is the dominant frequency of the acoustic pressure measurement from Task 1.

In the above equation (9), the value of τ should not be too small or too large because a small value τ would filter out more noise and a large value τ would include more information and data. Therefore, selecting the most accurate value for τ is necessary. By multiplying the filter with the Fourier transformed data of the acoustic pressure measurements, the noise would be removed and provide the denoised data for the clean path of the submarine.

3. Algorithm Implementation and Development

The algorithms that were used are Fourier transform (1) and Inverse Fourier transform (2). When computing the DFT, it causes $O(N^2)$ computational time which is expensive with big datasets. Fast Fourier Transform (FFT) and Inverse Fast Fourier Transform (iFFT) would be needed in this case to solve this problem because it computes the DFT and iDFT in a more efficient way by only causing $O(N \log N)$ computational time. The following illustrates when and how the FFT, iFFT, and Fourier shift (3) are used to identify the dominant frequency, filter out noisy data, and provide a clear path of the submarine.

The Fourier transform (1) is employed in Task 1 to obtain the dominant frequency. A for-loop iterates through the acoustic pressure data over time, and for each instance, the data undergoes a Fourier transformation for 3D (1) and then shifting (3). Within the loop, the Fourier transformed and shifted data accumulates. Once outside the loop, this accumulated data is divided by the total duration to provide the average of the Fourier transform data. Consequently, the index of the dominant frequency corresponds to the index of the maximum value of the average Fourier transform data. The Inverse Fourier transform (2) is utilized in Task 2 within the for-loop, after multiplying the filter by the Fourier transform of the original data at each time step. By inverse Fourier shifting (3) and inverse Fourier transforming (2) this data, it is returned to the signal space. This results in the denoised data for each time instance.

The software packages employed are NumPy, Plotly, and Matplotlib. NumPy [5] facilitates numerical computations such as Fast Fourier Transform and meshgrid operations. Plotly [7] and Matplotlib [6] provide capabilities for 2D and 3D plotting of the dominant frequency and path of the submarine.

4. Computational Results

Task 1:

Through averaging of signals after the Fourier transform, the dominant frequency that is generated by the submarine is determined. The index of the dominant frequency of the given data in Fourier space for (x, y, z) is $(39, 49, 10)$ and the dominant frequency is $(5.340707511102648, 2.199114857512855, -6.911503837897545)$.

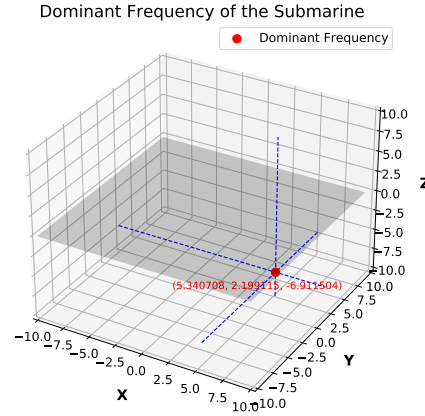


FIGURE 1. The dominant frequency in Fourier space in a 3D plot.

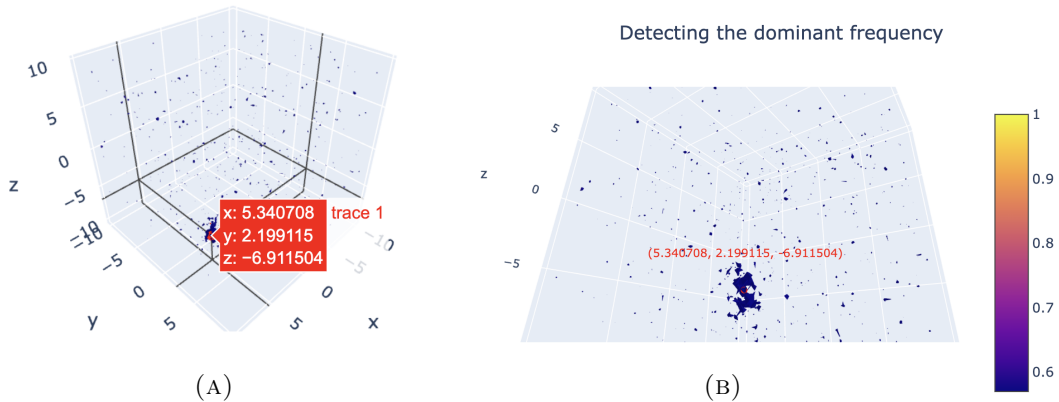


FIGURE 2. The normalization of the averaging of the Fourier transform data in a 3D plot.

Figure 2a and Figure 2b clarifies that (5.340707511102648, 2.199114857512855, -6.911503837897545) is the dominant frequency by normalization. The dominant frequency would be the highest number which is shown on Figure 2a. Figure 2a and Figure 2b shows the normalized averaging of the Fourier Transform data and the point that is the closest to 1 is (5.340708, 2.199115, -6.911504) which is the dominant frequency.

Task 2:

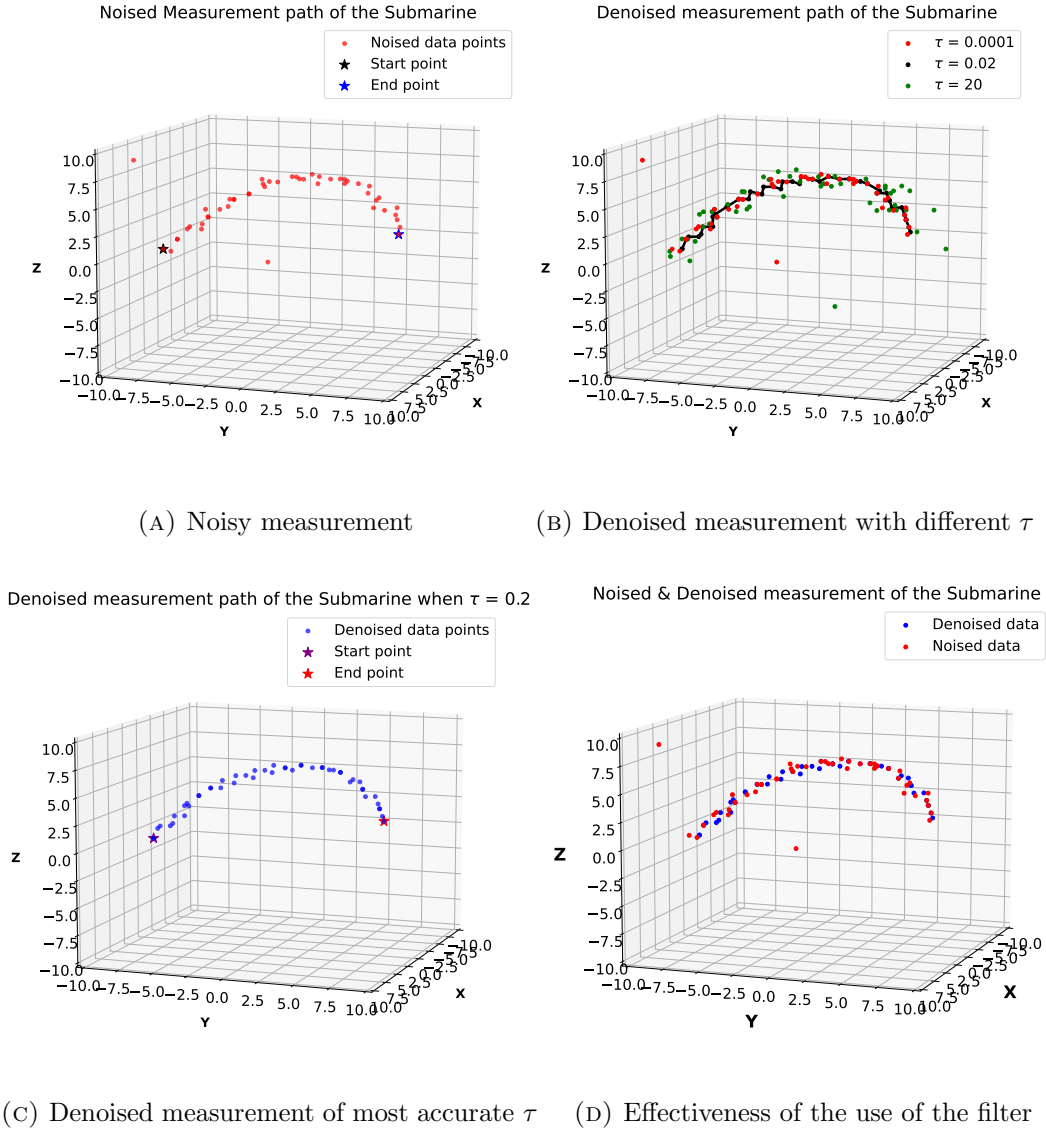


FIGURE 3. Plots after applying effective denoise filter to the noisy acoustics pressure data that leads to determining a more robust path for the submarine.

Figure 3a shows the path of the submarine in the acoustic pressure original data with noise. Figure 3b shows the path of the submarine that is filtered with three different τ values. When τ is either too small or too large, the filtering does not provide an accurate path and just causes confusion. Figure 3c uses $\tau = 0.2$ which is valid to provide the clearest path of the submarine. Figure 3d compares the noised and denoised measurement paths of the submarine for comparison. It shows that the filter effectively worked well.

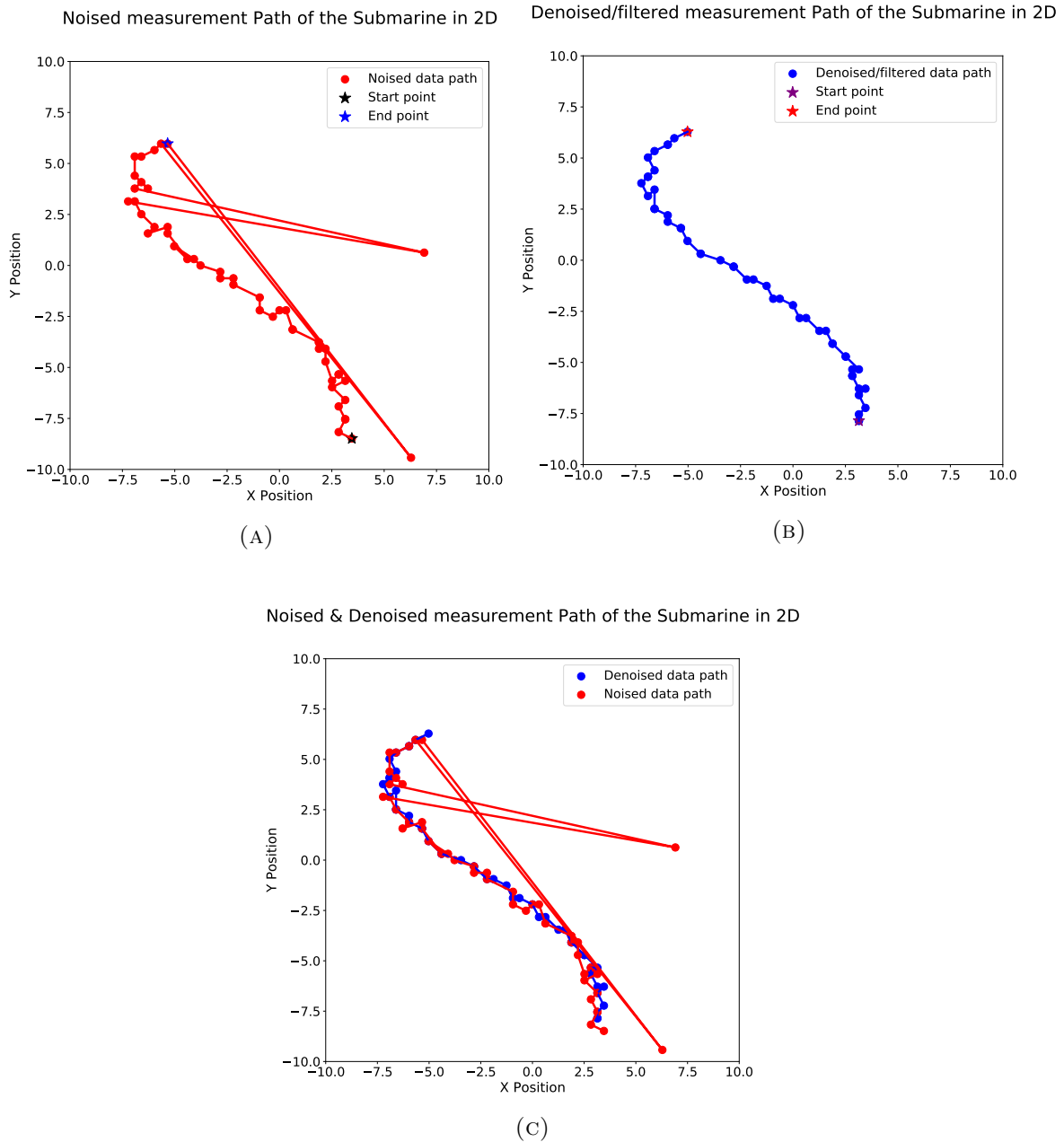
Task 3:

FIGURE 4. The (x, y) plots of the submarine path during the 24-hour period for both the noisy and denoised acoustic pressure measurements.

Figure 4a presents the noisy acoustic pressure measurement path of the submarine plotted in two dimensions. It seems like the submarine is moving in extreme dynamics back and fourth. Figure 4b shows the denoised acoustic pressure measurement path of the submarine plotted in two dimensions. It shows the submarine is moving in a clear and direct path. Figure 4c shows the comparison of how effectively the 3D Gaussian filter worked well for getting a clear and accurate path of the submarine.

5. Summary and Conclusions

The goal of this study was to detect the location of the submarine that is moving in the Puget Sound and emits unknown acoustic frequency. By using Fourier transform and 3D Gaussian filter to the collected signal data each time, the study was able to receive clean and clear data effectively that presents the path of the submarine in a 3D space. By utilizing the data obtained above and generating the corresponding 3D plot, the clean path of the submarine was identified. Application of these mathematical implementations develops this study to receive a valid result. Furthermore, this study can develop a system designed to detect submarines in noisy environments and establish a sub-tracking aircraft in the future, providing enhanced resources in acoustic technology and autonomous systems in marine engineering.

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